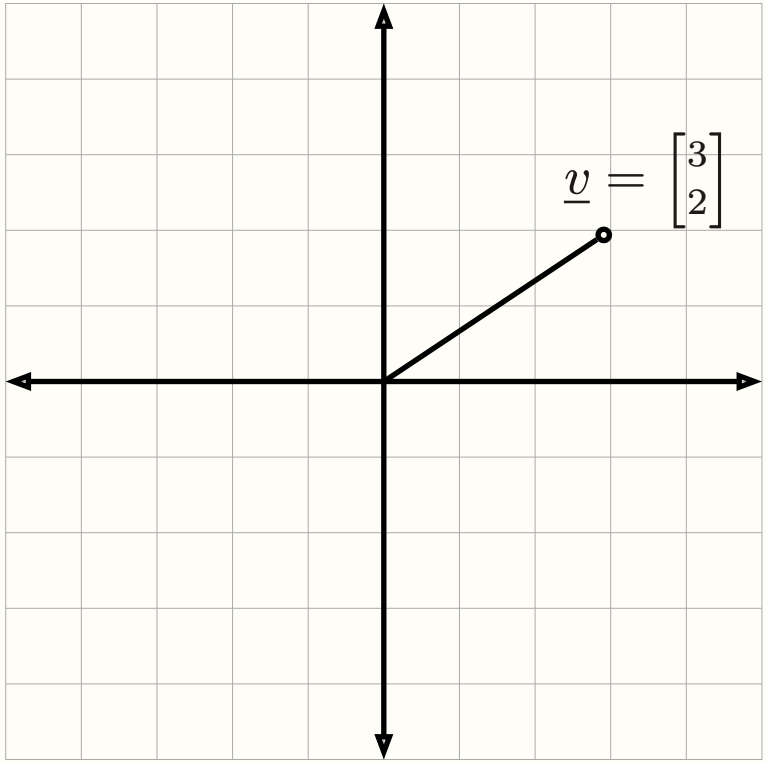
Linear Algebra II: geometry

Week 3 — Algorithmic Approaches to Mathematics

Dhruva Raman

Some recaps

Recap of norm



Calculate L_1,L_2,L_∞ norms of $\underline{v}$

Reminder

- $\|\underline{x}\|_2 = \sqrt{\sum(x_i^2)}$
- $\|\underline{x}\|_1 = \sum|x_i|$
- $\|\underline{x}\|_\infty = \max_i|x_|i|$

Last week's question

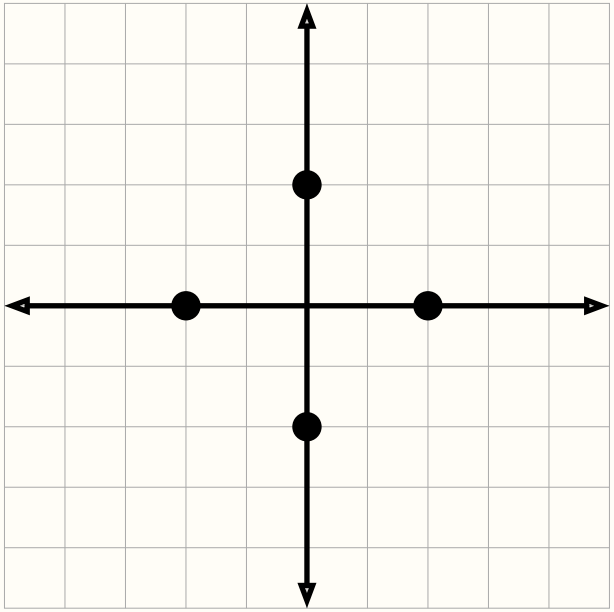
? For next week

- Draw the unit spheres in L_1, L_2, L_∞

$$S = \{\underline{v} \in \mathbb{R}^2 : \|\underline{v}\| = 1\}$$

Reminder

- $\|\underline{x}\|_2 = \sqrt{\sum(x_i^2)}$
- $\|\underline{x}\|_1 = \sum|x_i|$
- $\|\underline{x}\|_\infty = \max_i|x_i|$



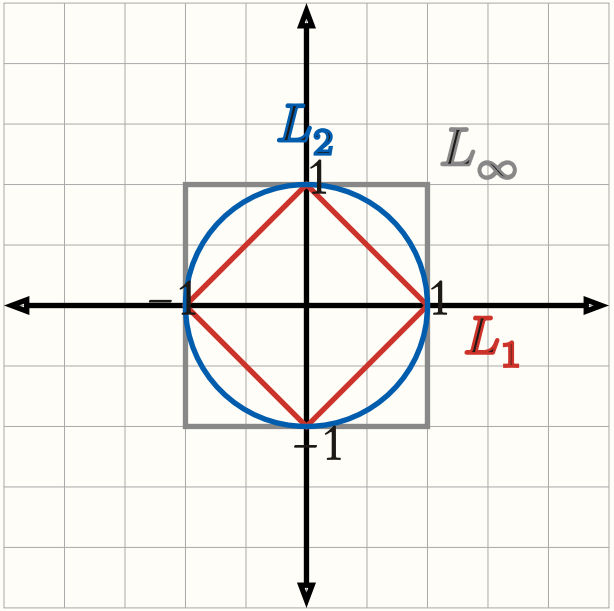
Hint: all marked points are on all spheres!

Last week's question

$$S_1 = \{\underline{x} \in \mathbb{R}^2 : |x_1| + |x_2| = 1\}$$

$$S_2 = \left\{\underline{x} \in \mathbb{R}^2 : \sqrt{x_1^2 + x_2^2} = 1\right\}$$

$$S_\infty = \left\{\underline{x} \in \mathbb{R}^2 : \max_i |x_i| = 1\right\}$$



Last week's movies

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	−1	−1
Bob	1	1	−1	−1
Carol	−1	−1	1	1
Eve	1	−1	0	0
Frank	10	10	−10	−10

Normalised[1]: $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

Normalised correlation: $\begin{bmatrix} 1 & 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 & 1 \\ -1 & -1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & -1 & 0 & 1 \end{bmatrix}$

[1] In $\mathcal{L}_2$, which is what people usually mean when not qualified

Last week's movies

Egregious hack

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	−1	−1
Bob	1	1	−1	−1
Carol	−1	−1	1	1
Eve	1	−1	0	0
Frank	10	10	−10	−10

Row means?

Last week's movies

Egregious hack

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	−1	−1
Bob	1	1	−1	−1
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Eve	1	−1	0	0
Frank	10	10	−10	−10

Row means?

Data is mean-centered.

What if scale was 1-10?

Last week's movies

Egregious hack

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Alice	1	1	−1	−1
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Frank	10	10	−10	−10

Always mean-center data for correlations!

Row means?

Data is mean-centered.

What if scale was 1-10?

No negative dot products possible!

Vector spaces

What really is a vector

$$\begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix} \in \mathbb{R}^3$$

- Images? Video/audio signals?
- Neural network weights?

What really is a vector

$$\begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix} \in \mathbb{R}^3$$

- Images? Video/audio signals?
- Neural network weights?

Vectors are objects that live in a vector space

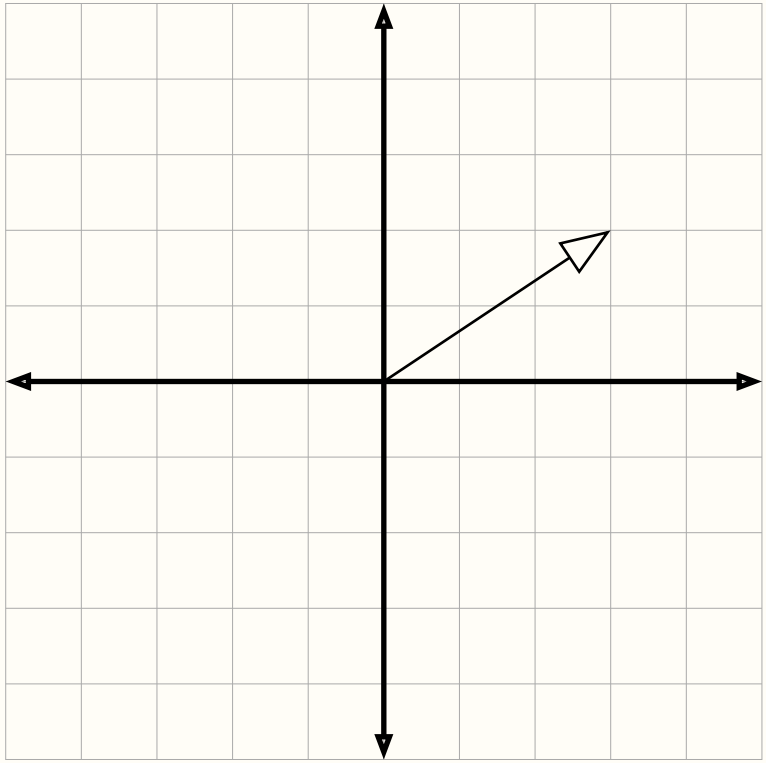
Goal

Define requirements on a vector space,

Hint

Vectors defined by how they interact, not by what they are

Vectors live in a vector space



$$\underline{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \in \mathbb{R}^2$$

$\mathbb{R}^2$ is a vector space

Other vector spaces

- $\mathbb{C}^3$
- $\mathbb{R}^{3\times 2}$ (yes matrices!)
- $\mathbb{R}^{3\times 2\times 4\times 5}$

- All are sets of shaped collections of numbers
- $\mathbb{C}$, $\mathbb{R}$ are the fields of the vector spaces

Vector spaces sit on a field K

What is a field?

Collection of scalars satisfying certain properties[1]

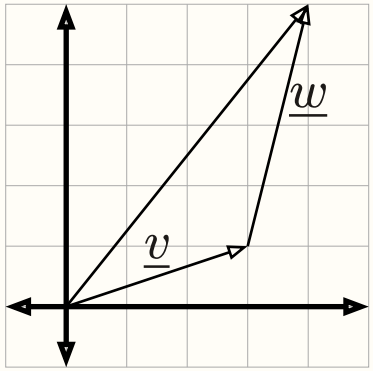
- The field is almost always the real numbers: $\mathbb{R}$
- Occasionally, it is $\mathbb{C}$
- $\mathbb{Z}$, $\mathbb{N}$, etc are not fields

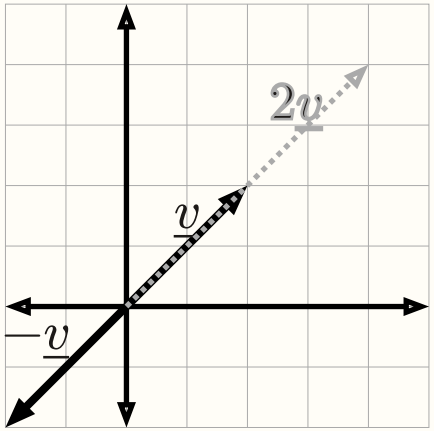
Take-home

Read $K = \mathbb{R}$

[1] see cheatsheet for these axioms

What makes a vector space V





Closure under addition[1]

$\forall \underline{v}, \underline{w} \in V : \underline{v} + \underline{w} \in V$

Closure under scalar multiplication

$\forall \underline{v} \in V, k \in K : k\underline{v} \in V$

Additive identity and inverse

$$\exists \underline{0} \in V : \underline{v} + \underline{0} = \underline{v} \quad \forall v \in V$$

$$\forall \underline{v} \in V \quad \exists -\underline{v} \in V : \underline{v} + -\underline{v} = \underline{0}$$

[1] Other more boring stuff in cheatsheet. No multiplication between vectors

What's the point of this abstraction

Tangible things (e.g. audio signals) are vector spaces

Mathematical machinery built for general vector spaces applies to specific ones

What vector spaces don't automatically give you[1]

- Angle (Inner product[2] space)
- Distance (Metric space)

…Let's think about these concepts now!

[1] but most have

[2] General name for dot product

Building vector space properties

Or “all about linearity”

The span of vectors: what can you reach?

Some vectors

$$\underline{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \underline{v}_2 = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$$

Can I make $\underline{v}_3 = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ by weighted addition of them?

The span of vectors: what can you reach?

Some vectors

$$\underline{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \underline{v}_2 = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$$

Can I make $\underline{v}_3 = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ by weighted addition of them?

Yes! $\underline{v}_3 = -2\underline{v}_1 + 1\underline{v}_2$

$\underline{v}_3$ is in the span of $\{\underline{v}_i\}_{i=1}^{2}$

Another vector

$$\underline{v}_4 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Cannot make $\underline{v}_4$ by weighted summation of $\{\underline{v}_i\}_{i=1}^{2}$

$\underline{v}_4$ is not in the span of $\{\underline{v}_i\}_{i=1}^{2}$

There are many people but only a few personalities

How many people are in the span of Alice alone?

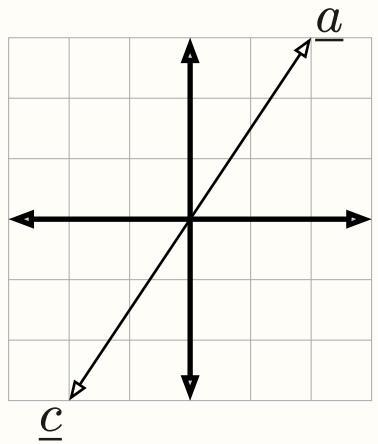
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[1] Always pretend vectors are in $\mathbb{R}^2$ and visualise!

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Frank	10	10	−10	−10



- $\underline{b} = \underline{a}$
- $\underline{c} = -\underline{a}$
- $\underline{f} = 10\underline{a}$

They are all parallel to $\underline{a}$[1]

[1] Always pretend vectors are in $\mathbb{R}^2$ and visualise!

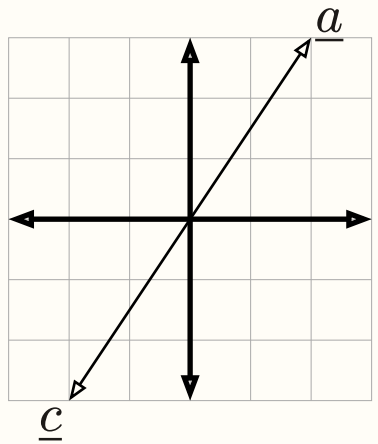
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- $\underline{b} = \underline{a}$
- $\underline{c} = -\underline{a}$
- $\underline{f} = 10\underline{a}$

They are all parallel to $\underline{a}$[1]



Exercise

Find row vector $\underline{w}^T$ with

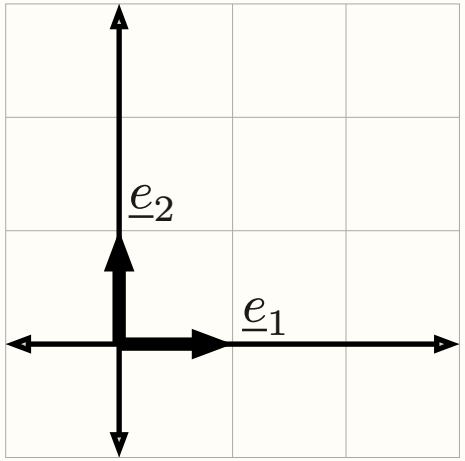
$$\text{hcat}\left(\underline{b}, \underline{c}, \underline{f}\right) = \underline{a}\underline{w}^T$$

[1] Always pretend vectors are in $\mathbb{R}^2$ and visualise!

A very special set of vectors

Some vectors

$$\underline{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \underline{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Same question, different words

Can you make any vector in $\mathbb{R}^2$ by weighted summation of these?

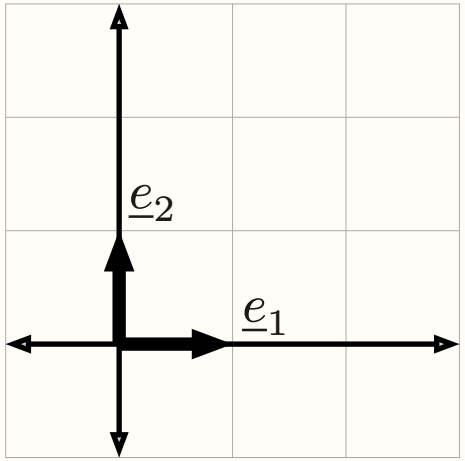
Do these vectors span $\mathbb{R}^2$?

[1] Notice difference between $\underline{v}_1$ (vector) and v_1 (element of vector $\underline{v}$)

A very special set of vectors

Some vectors

$$\underline{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \underline{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Same question, different words

Can you make any vector in $\mathbb{R}^2$ by weighted summation of these?

Do these vectors span $\mathbb{R}^2$?

Yes!

Let $\underline{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$.[1] Then

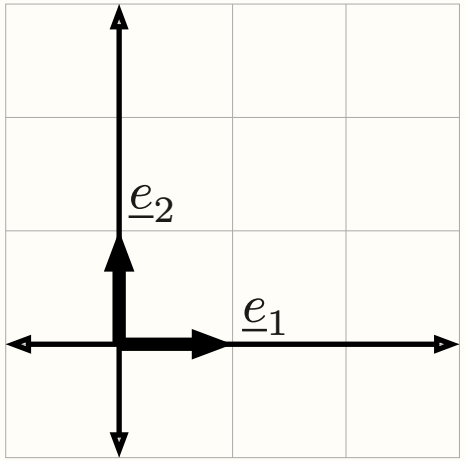
$$\underline{v} = v_1\underline{e}_1 + v_2\underline{e}_2$$

[1] Notice difference between $\underline{v}_1$ (vector) and v_1 (element of vector $\underline{v}$)

A very special set of vectors

Some vectors

$$\underline{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \underline{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Example

$$\underline{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

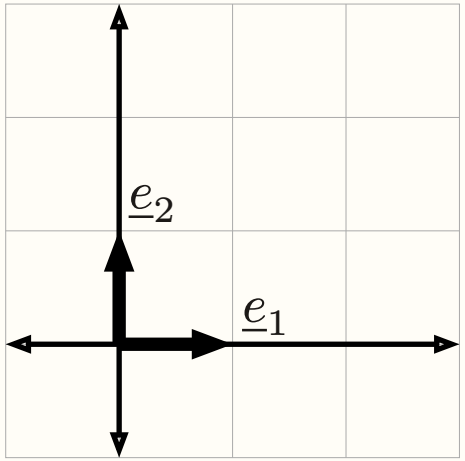
$$\underline{v} = 3\underline{e}_1 + 4\underline{e}_2$$

These vectors form a co-ordinate system. The standard (Cartesian) one.

A very special set of vectors

Some vectors

$$\underline{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \underline{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Example

$$\underline{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

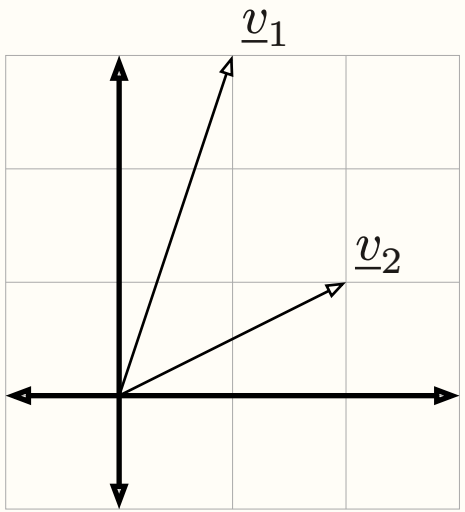
$$\underline{v} = 3\underline{e}_1 + 4\underline{e}_2$$

These vectors form a co-ordinate system. The standard (Cartesian) one.

A very special set of vectors

Some vectors

$$\underline{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \underline{v}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$



Do these span $\mathbb{R}^2$?

Trick

Can we make $\underline{e}_1$ and and $\underline{e}_2$ from them?

$$\underline{e}_1 = -\tfrac{1}{5}\underline{v}_1 + \tfrac{3}{5}\underline{v}_2$$

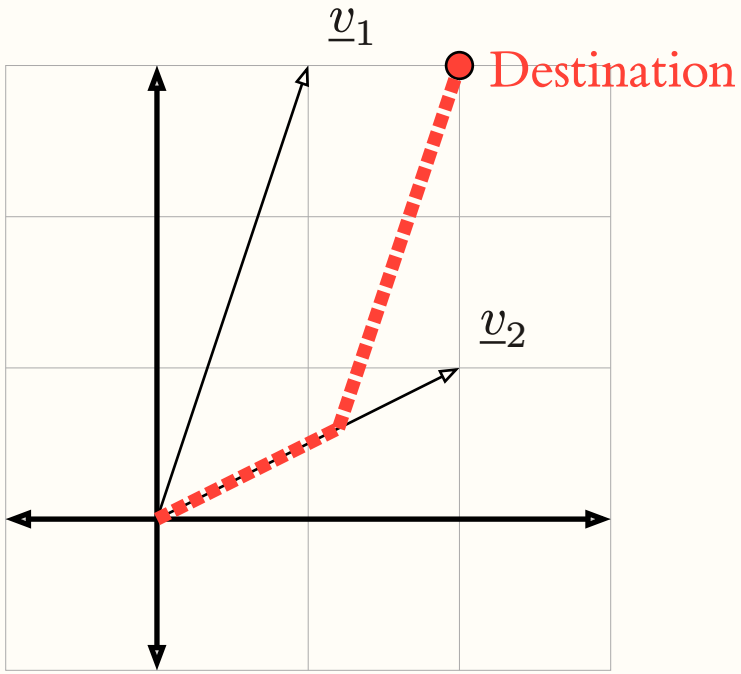
$$\underline{e}_2 = \tfrac{2}{5}\underline{v}_1 - \tfrac{1}{5}\underline{v}_2$$

Exercise

Build matrix that turns $\text{hcat}(\underline{v}_1, \underline{v}_2)$ into $\text{hcat}(\underline{e}_1, \underline{e}_2)$

A very special set of vectors

Some vectors



Trick, restated

$$\underline{v}_1, \underline{v}_2 \underbrace{\rightarrow}_{\text{convert into}} \underline{e}_1, \underline{e}_2 \underbrace{\rightarrow}_{\text{convert into}} \text{anything}$$

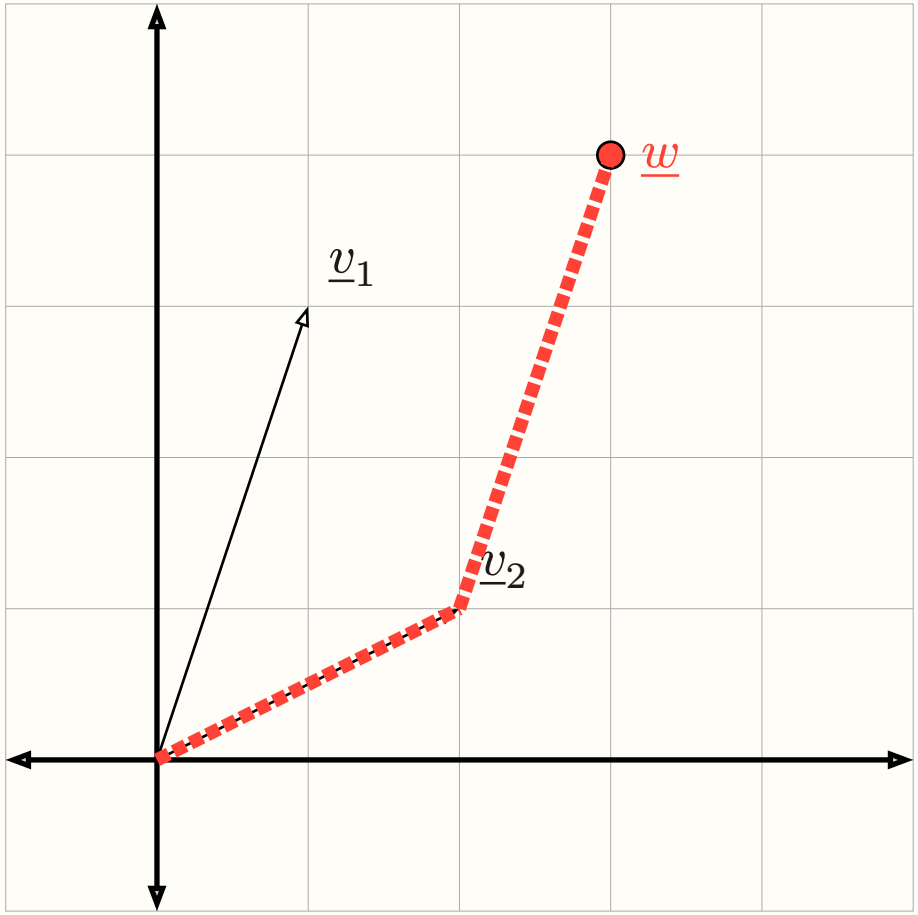
When does this trick work?

- Always if $\underline{v}_1$ and $\underline{v}_2$, are not parallel

Intuition

Drive from zero, only in $\pm$ these directions.

A different co-ordinate system



In the co-ordinate system of $\underline{v}_1$ and $\underline{v}_2$:

$$\underline{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

In the Cartesian co-ordinate system of $\underline{e}_1$ and $\underline{e}_2$:

$$\underline{w} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Exercise for home

Let $S = \text{span}(\{\underline{v}_1, \underline{v}_2, ..., \underline{v}_n\})$ where $\underline{v}_i \in V$

- Show that S is itself a vector space inside V
- This is called a (vector) subspace

Linear independence

Some linearly dependent vectors

$$\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 9 \\ 7 \\ 1 \end{bmatrix}$$

Linear independence

Some linearly dependent vectors

$$\begin{bmatrix}1\\3\\1\end{bmatrix}, \begin{bmatrix}4\\2\\0\end{bmatrix}, \begin{bmatrix}9\\7\\1\end{bmatrix}$$

- $1\underline{v}_1 + 2\underline{v}_2 = \underline{v}_3$
- $1\underline{v}_1 + 2\underline{v}_2 - \underline{v}_3 = \underline{0}$ (equivalent)

$\underline{v}_3$ is in span of the others

Linear independence

Some linearly dependent vectors

$$\begin{bmatrix}1\\3\\1\end{bmatrix}, \begin{bmatrix}4\\2\\0\end{bmatrix}, \begin{bmatrix}9\\7\\1\end{bmatrix}$$

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Some linearly independent vectors

$$\begin{bmatrix}1\\3\\1\end{bmatrix}, \begin{bmatrix}4\\2\\0\end{bmatrix}, \begin{bmatrix}9\\7\\12\end{bmatrix}$$

Linear independence

Some linearly dependent vectors

$$\begin{bmatrix}1\\3\\1\end{bmatrix}, \begin{bmatrix}4\\2\\0\end{bmatrix}, \begin{bmatrix}9\\7\\1\end{bmatrix}$$

- $1\underline{v}_1 + 2\underline{v}_2 = \underline{v}_3$
- $1\underline{v}_1 + 2\underline{v}_2 - \underline{v}_3 = \underline{0}$ (equivalent)

$\underline{v}_3$ is in span of the others

Some linearly independent vectors

$$\begin{bmatrix}1\\3\\1\end{bmatrix}, \begin{bmatrix}4\\2\\0\end{bmatrix}, \begin{bmatrix}9\\7\\12\end{bmatrix}$$

- Can't make one from others!
- Could I add a fourth independent vector?

Nobody in span of anybody else

Some linearly dependent vectors

$$\begin{bmatrix}1\\3\\1\end{bmatrix}, \begin{bmatrix}4\\2\\0\end{bmatrix}, \begin{bmatrix}9\\7\\1\end{bmatrix}$$

- $1\underline{v}_1 + 2\underline{v}_2 = \underline{v}_3$
- $1\underline{v}_1 + 2\underline{v}_2 - \underline{v}_3 = \underline{0}$ (equivalent)

Can express one as weighted sum of others

Formal definition of linear dependence

$\{\underline{v} \in V\}_{i=1}^{n}$ are linearly dependent if $\exists\{c_i \in K\}_{i=1}^{n}$, not all zero, such that

$$\sum_{i=1}^{n} c_i \underline{v}_i = \underline{0}$$

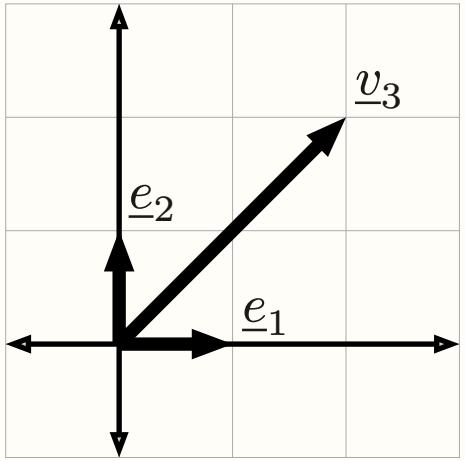
Back to movies

	Dune	Interstellar	Love actually	Mamma Mia
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Carol	−1	−1	1	1
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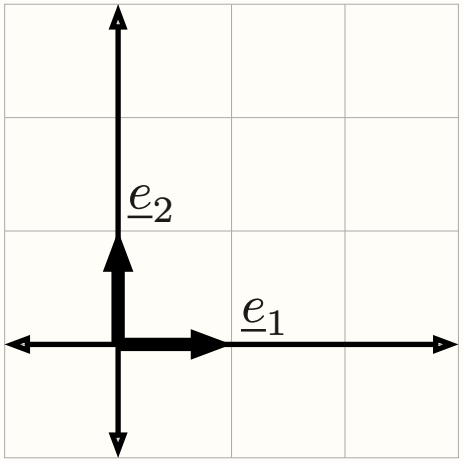
Biggest set of linearly independent people you can make?

Linearly dependent vectors are redundant

Bases: reach everything, waste nothing



- Span entire vector space
- Redundant (linearly dependent)



- Minimal spanning set

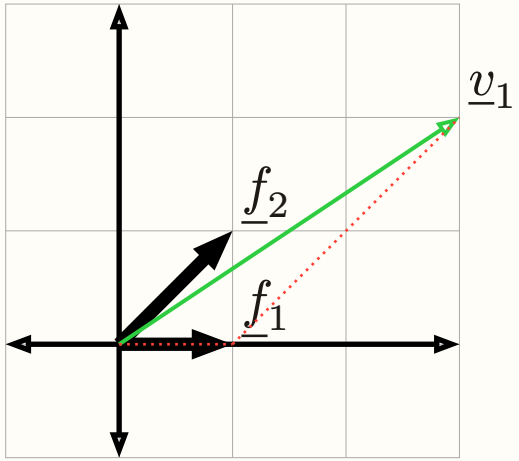
… is called a basis of the vector space[1]

[1] and $\underline{e}_1$, $\underline{e}_2$, is the *canonical* basis

A basis is a co-ordinate system

Is this a basis?

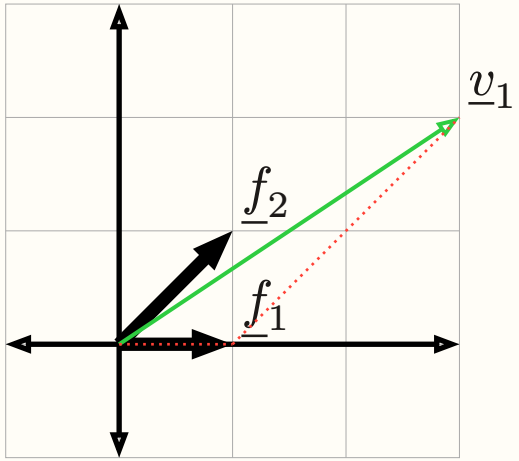
$$\underline{f}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \underline{f}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



A basis is a co-ordinate system

Is this a basis?

$$\underline{f}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \underline{f}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$\underline{v}_1$ **in canonical basis:**

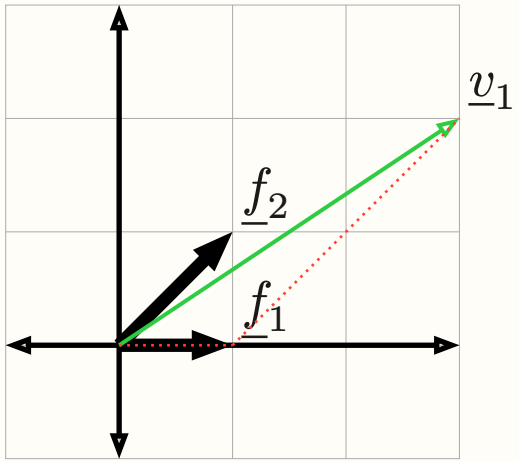
$$\underline{v}_1 = 3\underline{e}_1 + 2\underline{e}_2$$

- Represented as $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

A basis is a co-ordinate system

Is this a basis?

$$\underline{f}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \underline{f}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$\underline{v}_1$ **in canonical basis:**

$$\underline{v}_1 = 3\underline{e}_1 + 2\underline{e}_2$$

- Represented as $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

$\underline{v}_1$ **in $\underline{f}$ basis:**

$$\underline{v}_1 = 1\underline{f}_1 + 2\underline{f}_2$$

- Represented as $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

- Same vector (geometric)
- Different representation

Interlude: a new way of multiplying matrices

Old

$$\begin{bmatrix} \text{prices} \rightarrow \\ \text{weights} \rightarrow \end{bmatrix} \begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} 🍏 \\ 🍌 \end{bmatrix}$$

$$\begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix} \left(\begin{bmatrix} £ \\ \text{kg} \end{bmatrix} \right)$$

Rows on left ask questions

- price/weight of all items

Columns on right answer

- Per-basket

Interlude: a new way of multiplying matrices

Old

$$\begin{bmatrix}\text{prices} \to \\ \text{weights} \to\end{bmatrix} \left[\begin{array}{cc}1.50 & 0.75 \\ \hdashline 0.5 & 0.25\end{array}\right] \begin{bmatrix}🍏 \\ 🍌\end{bmatrix}$$

$$\left[\begin{array}{cc}1.50 & 0.75 \\ \hdashline 0.5 & 0.25\end{array}\right] \begin{bmatrix}3 \\ 2\end{bmatrix} = \begin{bmatrix}6 \\ 2\end{bmatrix} \left(\begin{bmatrix}£ \\ \text{kg}\end{bmatrix} \right)$$

Rows on left ask questions

- price/weight of all items

Columns on right answer

- Per-basket

New

$$\left[\begin{array}{c|c}1.50 & 0.75 \\ 0.5 & 0.25\end{array}\right] \begin{bmatrix}3 \\ 2\end{bmatrix} = 3\begin{bmatrix}1.50 \\ 0.5\end{bmatrix} + 2\begin{bmatrix}0.75 \\ 0.25\end{bmatrix}$$

Rows on right ask questions

- How many of each fruit?

Columns on left answer

- All per-fruit data

Interlude: a new way of multiplying matrices

New

$$\left[\begin{array}{c|c} 1.50 & 0.75 \\ 0.5 & 0.25 \end{array}\right] \begin{bmatrix} 🍏 \\ 🍌 \end{bmatrix} = 🍏 \underbrace{\begin{bmatrix} 1.50 \\ 0.5 \end{bmatrix}}_{\underline{v}_1} + 🍌 \underbrace{\begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix}}_{\underline{v}_2}$$

Rows on right ask questions

- How many of each fruit?

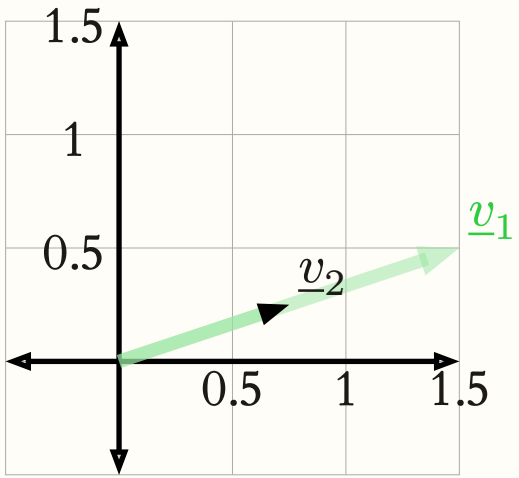
Columns on left answer

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Interlude: a new way of multiplying matrices

New

$$\left[\begin{array}{c|c} 1.50 & 0.75 \\ 0.5 & 0.25 \end{array}\right] \begin{bmatrix} 🍎 \\ 🍌 \end{bmatrix} = 🍎 \underbrace{\begin{bmatrix} 1.50 \\ 0.5 \end{bmatrix}}_{\underline{v}_1} + 🍌 \underbrace{\begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix}}_{\underline{v}_2}$$



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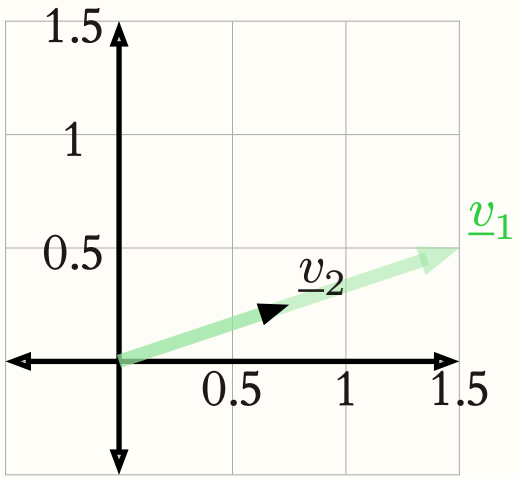
- All per-fruit data

Solve $\left[\begin{array}{c|c} 1.50 & 0.75 \\ 0.5 & 0.25 \end{array}\right] = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$?

Interlude: a new way of multiplying matrices

New

$$\left[\begin{array}{c|c} 1.50 & 0.75 \\ 0.5 & 0.25 \end{array}\right] \begin{bmatrix} 🍎 \\ 🍌 \end{bmatrix} = 🍎 \underbrace{\begin{bmatrix} 1.50 \\ 0.5 \end{bmatrix}}_{\underline{v}_1} + 🍌 \underbrace{\begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix}}_{\underline{v}_2}$$



Rows on right ask questions

- How many of each fruit?

Columns on left answer

- All per-fruit data

Solve $\left[\begin{array}{c|c} 1.50 & 0.75 \\ 0.5 & 0.25 \end{array}\right] = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$?

Impossible!

Solution must be parallel to $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$!

Practice

$$\left[\begin{array}{c:c} 2 & 3 \\ 1 & 1 \\ 4 & 2 \end{array}\right] \begin{bmatrix} 4 & 2 \\ 6 & 2 \end{bmatrix}$$

Interlude: a new way of multiplying matrices

Practice

$$\left[\begin{array}{c:c} 2 & 3 \\ 1 & 1 \\ 4 & 2 \end{array}\right] \begin{bmatrix} 4 & 2 \\ 6 & 2 \end{bmatrix}$$

Left column (Alice's 4 🍏, 6 🍌 basket):

$$4\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} + 6\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

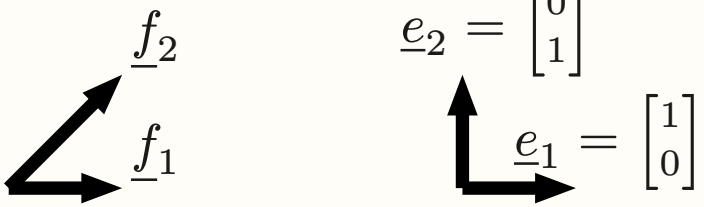
Right column (Bob's 2 🍏, 2 🍌 basket):

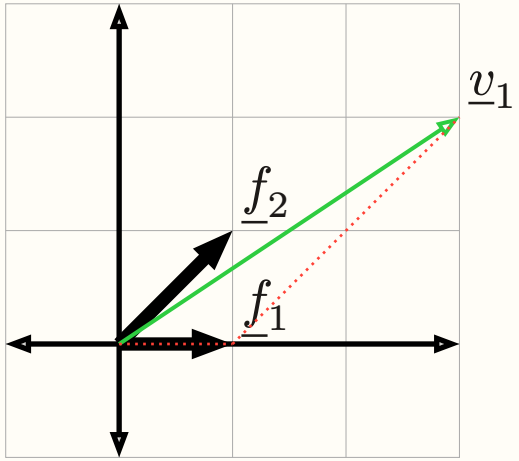
$$2\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} + 2\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

Result

$$\begin{bmatrix} 26 & 10 \\ 10 & 4 \\ 28 & 12 \end{bmatrix}$$

A basis is a co-ordinate system





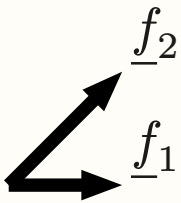
$$\underline{v} = 1\underline{f}_1 + 2\underline{f}_2$$

$$\underline{v} = 3\underline{e}_1 + 2\underline{e}_2$$

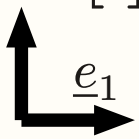
Can we convert from $\underline{f}$-basis into $\underline{e}$-basis?

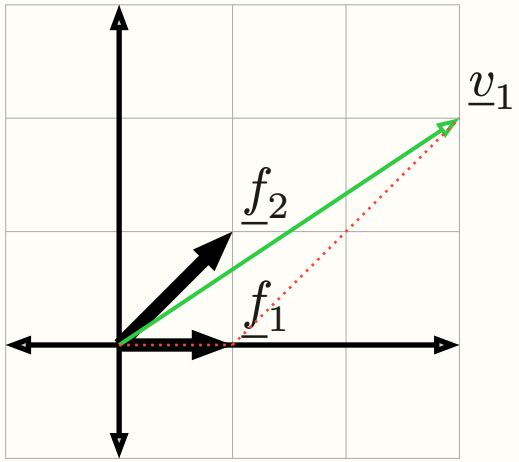
$$\underline{f}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \underline{f}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ in } \underline{e} \text{ basis}$$

A basis is a co-ordinate system



$$\underline{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$





$$\underline{v} = 1\underline{f}_1 + 2\underline{f}_2$$

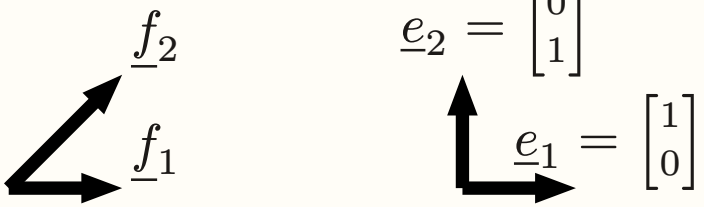
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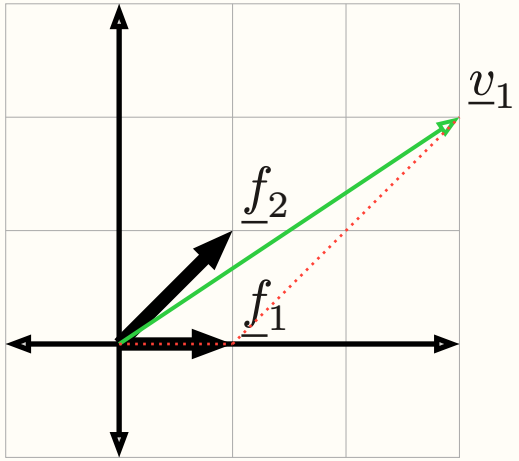
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$$\underline{v} = 1\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

A basis is a co-ordinate system





$$\underline{v} = 1\underline{f}_1 + 2\underline{f}_2$$

$$\underline{v} = 3\underline{e}_1 + 2\underline{e}_2$$

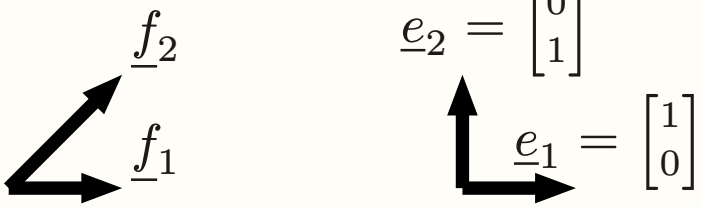
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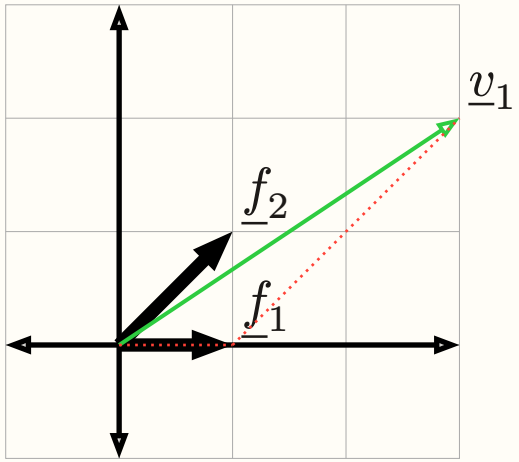
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$$= \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}$$

A basis is a co-ordinate system





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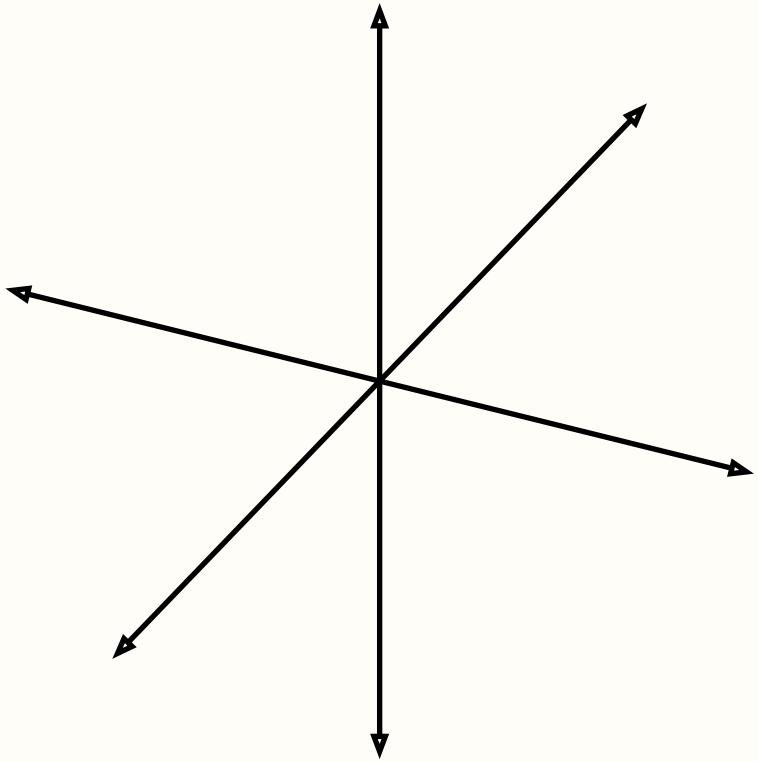
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$$= \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}$$

Exercises

- Verify matrix works for any $v \in \mathbb{R}^2$
- Find matrix converting back from $\underline{e}$ to $\underline{f}$ basis. Is it inverse?

Defining the dimension of a vector space



$$\underline{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \underline{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \underline{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

It's the length of a basis of the vector space

Key ingredient

- All bases have the same length!

Extension exercise

- Prove this
  (e.g. by contradiction)

Recap

$$\left[\begin{array}{c|c} 1.50 & 0.75 \\ 0.5 & 0.25 \end{array}\right] \begin{bmatrix} 🍏 \\ 🍌 \end{bmatrix} = 🍏 \underbrace{\begin{bmatrix} 1.50 \\ 0.5 \end{bmatrix}}_{\underline{v}_1} + 🍌 \underbrace{\begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix}}_{\underline{v}_2}$$

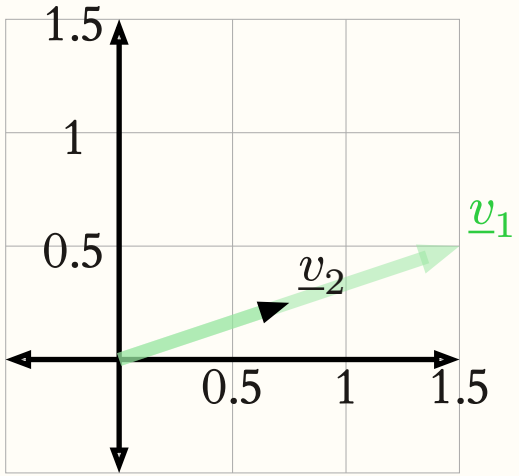


Image of the matrix

- is the set of possible outputs

$$\mathrm{Im}(A) = \mathrm{span}(\underline{v}_1)$$

[1] always: prove as an exercise

Recap

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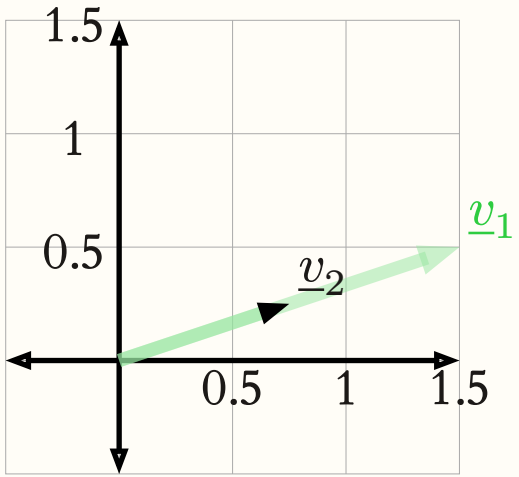


Image of the matrix

- is the set of possible outputs

$$\mathrm{Im}(A) = \mathrm{span}(\underline{v}_1)$$

- is a vector space![1] Dimension? Basis?

Rank of a matrix

$$\mathrm{rank}(A) = \dim(\mathrm{Im}(A))$$

[1] always: prove as an exercise

Recap

$$\left[\begin{array}{c|c} 1.50 & 0.75 \\ 0.5 & 0.25 \end{array}\right] \begin{bmatrix} 🍏 \\ 🍌 \end{bmatrix} = 🍏 \underbrace{\begin{bmatrix} 1.50 \\ 0.5 \end{bmatrix}}_{\underline{v}_1} + 🍌 \underbrace{\begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix}}_{\underline{v}_2}$$

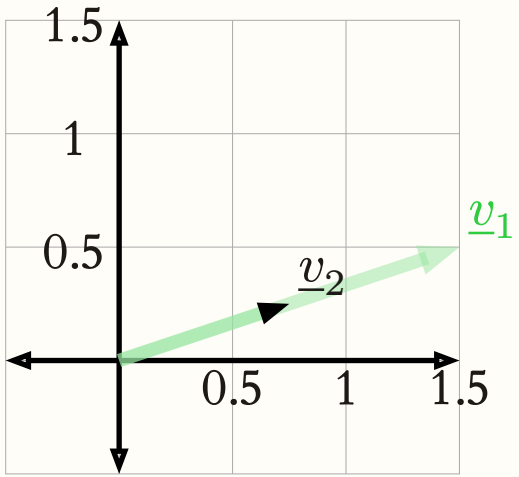


Image of the matrix

- is the set of possible outputs

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Rank of a matrix

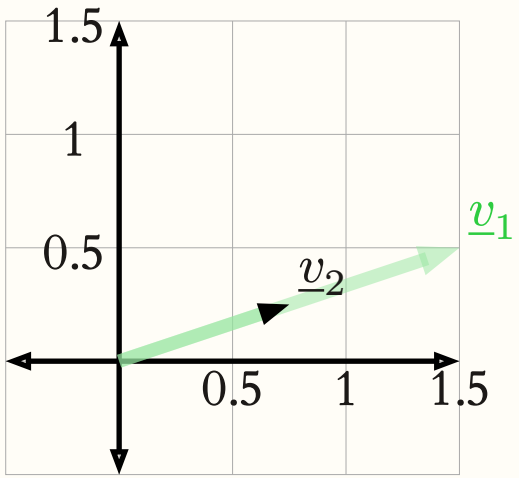
$$\mathrm{rank}(A) = \dim(\mathrm{Im}(A))$$

- This is a rank-1 matrix

[1] always: prove as an exercise

An equivalent definition

$$\left[\begin{array}{c|c} 1.50 & 0.75 \\ 0.5 & 0.25 \end{array}\right] \begin{bmatrix} 🍏 \\ 🍌 \end{bmatrix} = 🍏 \underbrace{\begin{bmatrix} 1.50 \\ 0.5 \end{bmatrix}}_{\underline{v}_1} + 🍌 \underbrace{\begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix}}_{\underline{v}_2}$$



Rank of a matrix

- Number of linearly independent columns[1]

Exercises

- Figure out why!
  (for square matrices)
- **Hard**: Prove (for all matrices)

  row rank = column rank

[1] This is called the column rank

There are many people but only two personalities

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	−1	−1
Bob	1	1	−1	−1
Carol	−1	−1	1	1
Eve	1	−1	0	0
Frank	10	10	−10	−10

Row rank? Column rank?

There are many people but only two personalities

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Row rank? Column rank?

It's a 2×2 matrix on stilts!

$$\begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 10 & 10 & -10 & -10 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 0 \\ 1 \end{bmatrix}$$ meaning?

There are many people but only two personalities

	Dune	Interstellar	Love actually	Mamma Mia
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How much people would like a new movie

Two people give all info!

Interim summary

The span of vectors forms a vector space V

Linearly independent spanning vectors form a basis of V

The number of basis elements is the dimension of V

Interim summary

The span of vectors forms a vector space V

Linearly independent spanning vectors form a basis of V

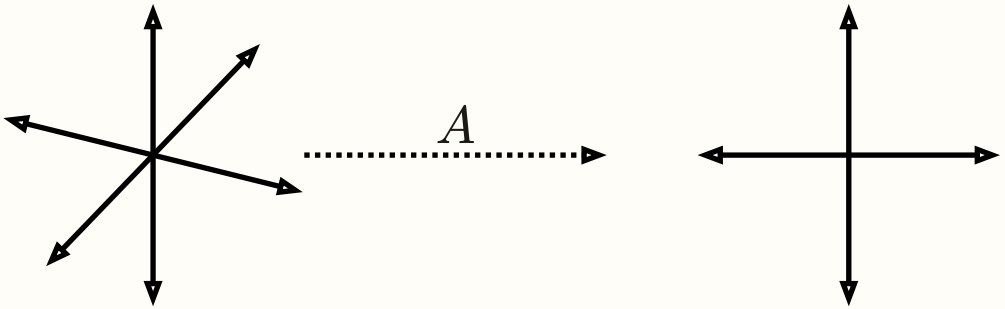
The number of basis elements is the dimension of V

The columns of a matrix M span $\mathrm{Im}(M)$ (thus a subspace)

The rank of a matrix is the dimension of $\mathrm{Im}(M)$

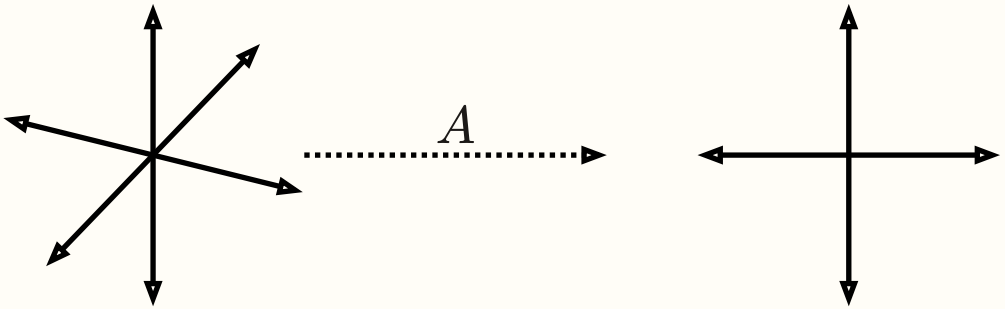
Rank-deficient matrices have linearly dependent columns

Rank-deficient (singular) matrices



$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \\ 0 & 0 & 0 \end{bmatrix} \quad A\underline{x} = \begin{bmatrix} \bullet \\ \bullet \\ 0 \end{bmatrix}$$

Rank-deficient (singular) matrices

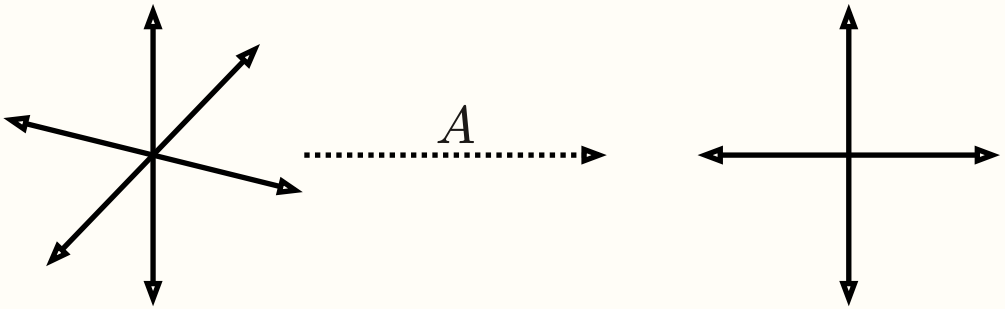


$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \\ 0 & 0 & 0 \end{bmatrix} \quad A\underline{x} = \begin{bmatrix} \bullet \\ \bullet \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \\ 4 & 8 & 0 \end{bmatrix}$$

Image of A is clearly a two-dimensional vector space

Rank-deficient (singular) matrices



$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \\ 0 & 0 & 0 \end{bmatrix} \quad A\underline{x} = \begin{bmatrix} \bullet \\ \bullet \\ 0 \end{bmatrix}$$

Image of A is clearly a two-dimensional vector space

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \\ 4 & 8 & 0 \end{bmatrix}$$

$$A\underline{x} = (x_1 + 2x_2) \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$$

Image of A is also a two-dimensional vector space

These are both Rank-2 matrices

Rank-deficient (singular) matrices

Exercise

Find a row vector $\underline{w}^T$ with

$$\text{hcat}\left(\underline{b}, \underline{c},, \underline{f}\right) = \underline{a}\underline{w}^T$$

Rank-deficient (singular) matrices

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \\ 4 & 8 & 0 \end{bmatrix}$$

- Invertible?

$$A\underline{x} = (x_1 + 2x_2)\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$$

- $\mathfrak{I}(A)$ is spanned by two vectors.
- It's a two-dimensional vector space.

Suppose it is invertible[1]...

$$\begin{aligned} \underline{x} &= A^{-1}A\underline{x} && \forall x \in \mathbb{R}^3 \\ &= A^{-1}\underline{y} && \text{where } \underline{y} \in \mathfrak{I}(A) \end{aligned}$$

[1] proof by contradiction incoming!

Rank-deficient (singular) matrices

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So

$$\underline{x} = w_1 A^{-1}\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + w_2 A^{-1}\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$$

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Rank-deficient (singular) matrices

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \\ 4 & 8 & 0 \end{bmatrix}$$

- Invertible?

$$A\underline{x} = (x_1 + 2x_2)\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$$

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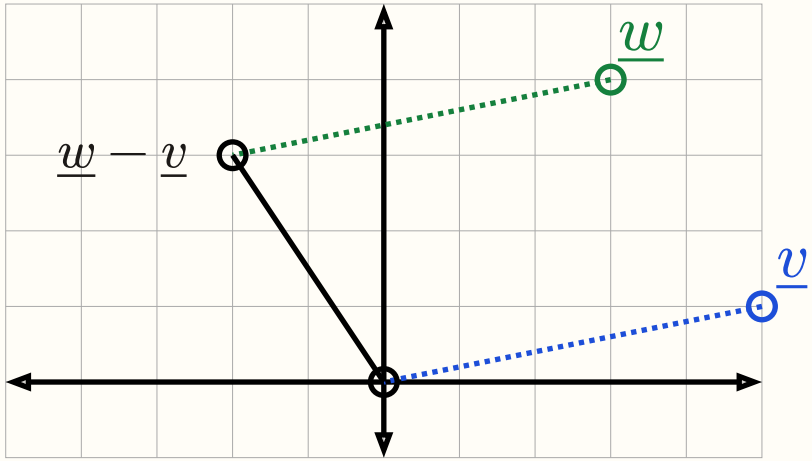
$$\underline{x} = w_1 A^{-1}\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + w_2 A^{-1}\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$$

Arbitrary $\underline{x}$ so $\mathbb{R}^3$ is spanned by two vectors! Contradiction

[1] proof by contradiction incoming!

Geometry of the dot product

Metrics: distances between vectors



Key point

Translating vectors equally doesn't change relative distance

Norm (recap)

- Notion of distance from 0

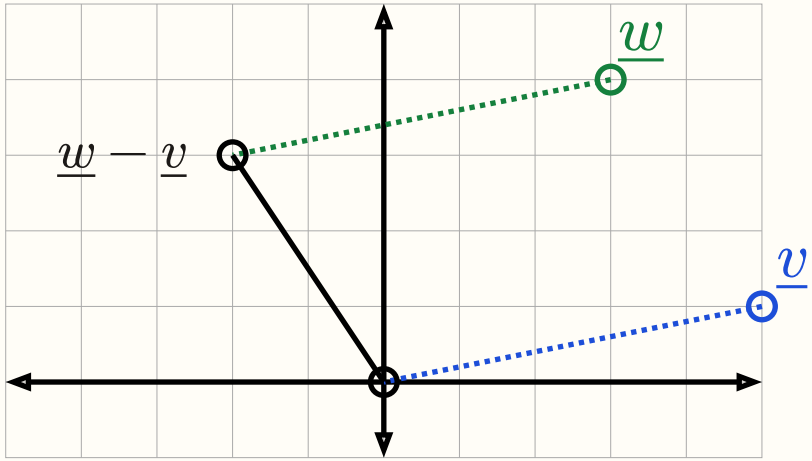
Metric

- Distance between vectors

Challenge

- Define metric from norm using graph on left?

Metrics: distances between vectors



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Metric

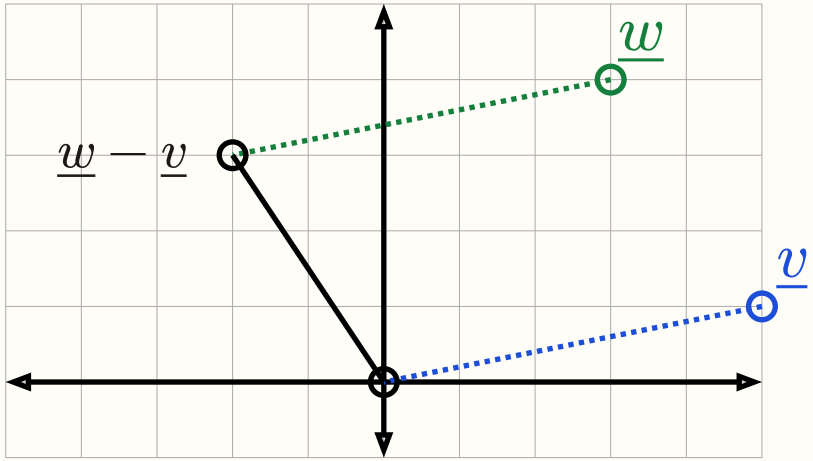
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Challenge

- Define metric from norm using graph on left?

$$d(\underline{v}, \underline{w}) = \|\underline{v} - \underline{w}\|$$

Metrics: distances between vectors



Each norm induces a metric

Example: $\mathcal{L}_2$ metric

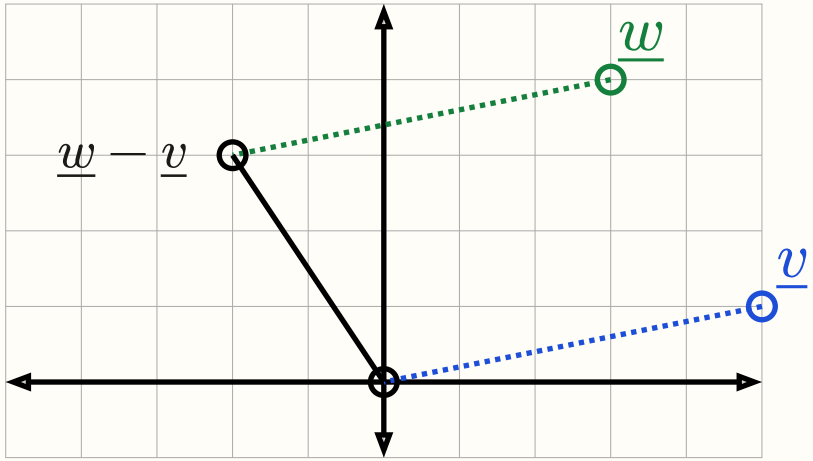
$$d(\underline{v}, \underline{w}) = \|\underline{v} - \underline{w}\|_2$$

Maths definition of metric

$$d : V \times V \to \mathbb{R}^{\geq 0}$$

- satisfying what properties?

Metrics: distances between vectors



Each norm induces a metric

Example: $\mathcal{L}_2$ metric

$$d(\underline{v}, \underline{w}) = \|\underline{v} - \underline{w}\|_2$$

Maths definition of metric

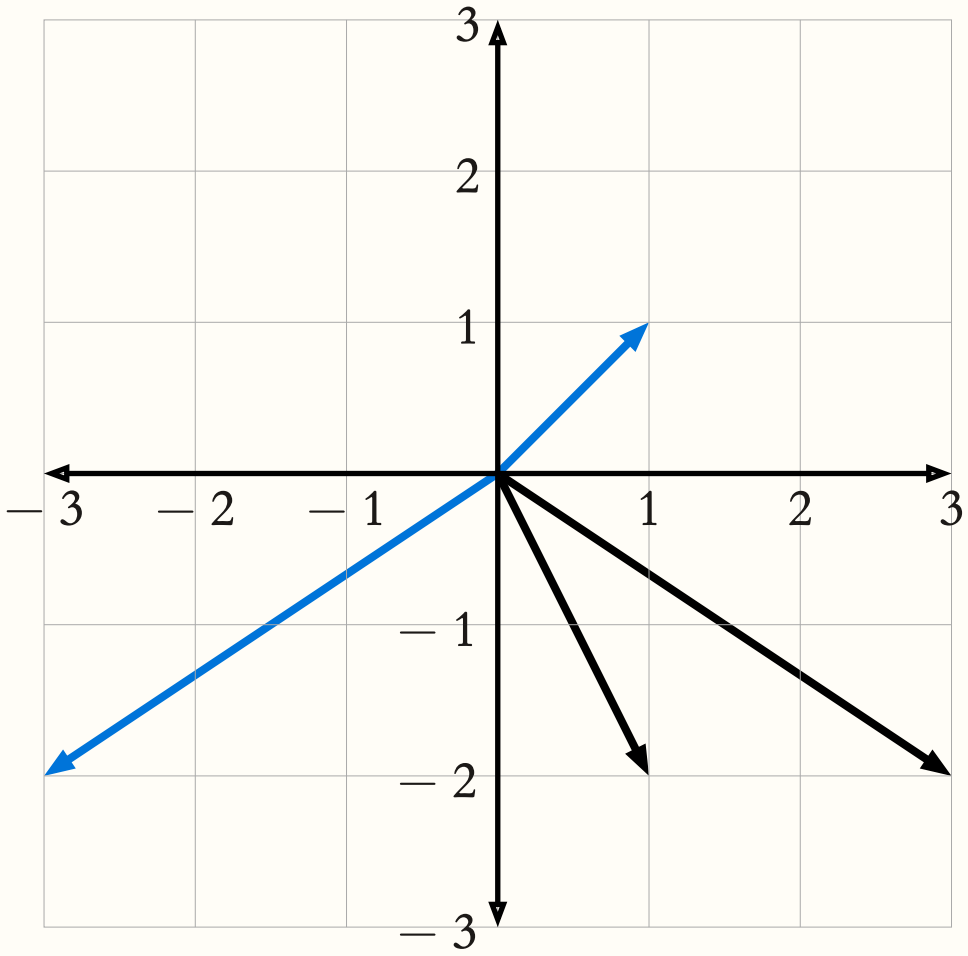
$$d : V \times V \to \mathbb{R}^{\geq 0}$$

- satisfying what properties?

Exercise:

Derive triangle inequality for metrics from triangle inequality for norms from last lecture

Dot products seem a bit geometric?



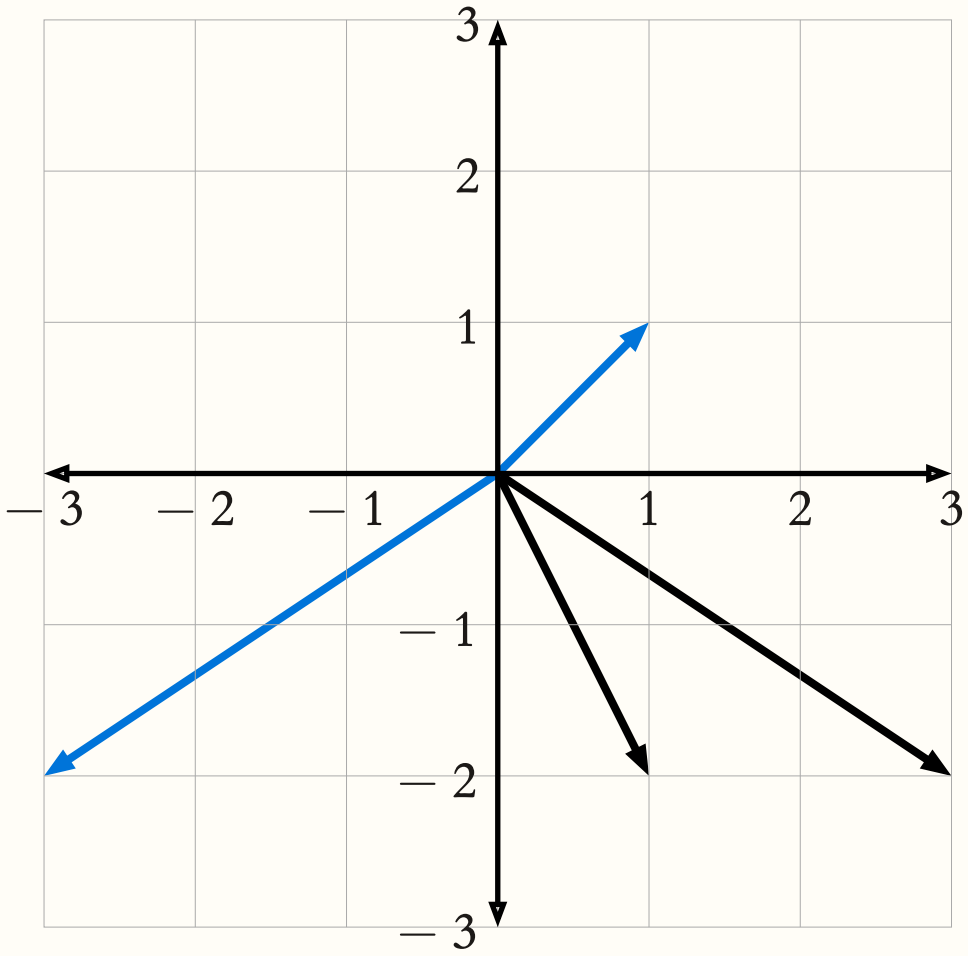
Pair in same quadrant

$\Rightarrow$ Sign of elements align

$\Rightarrow$ small angle between them

$\Rightarrow$ positive dot product

Dot products seem a bit geometric?



Pair in same quadrant

⇒ Sign of elements align

⇒ small angle between them

⇒ positive dot product

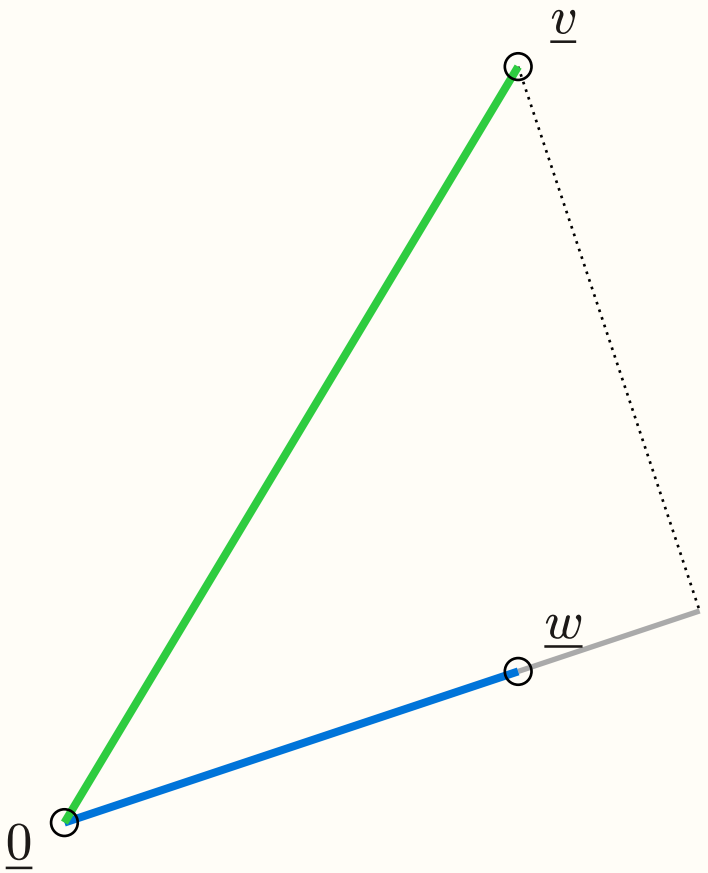
Pair in different quadrant

⇒ Sign of elements don't align

⇒ large angle between them

⇒ negative dot product

Dot products as projections

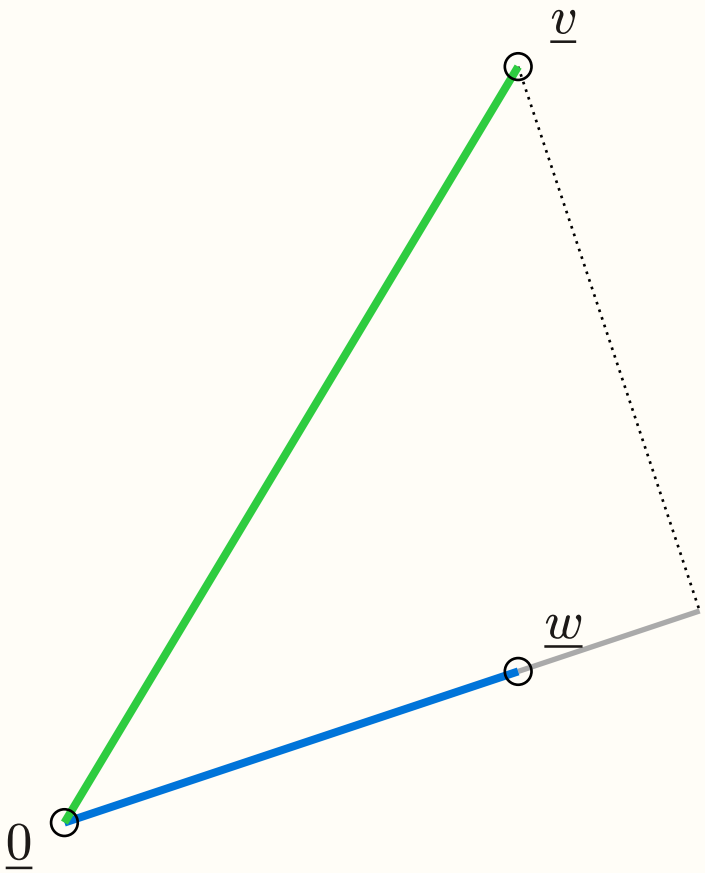


Geometric definition of dot product

- How far does the shadow of v extend along w?
- $\langle \underline{v}, \underline{w} \rangle = \|\underline{w}\| \times \|\text{shadow}\|$

[1] remember that mathematicians irrationally hate the $\times$ sign!

Dot products as projections



Geometric definition of dot product

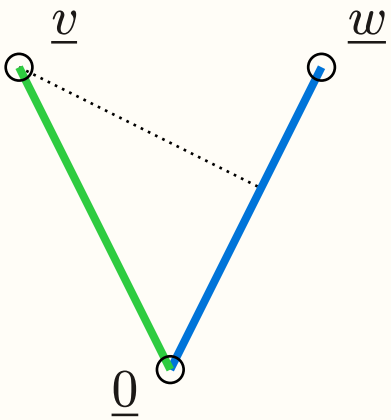
- How far does the shadow of v extend along w?
- $\langle \underline{v}, \underline{w} \rangle = \|\underline{w}\| \times \|\text{shadow}\|$

$$\langle \underline{v}, \underline{w} \rangle \approx \|\underline{w}\| \times 1.4\|\underline{w}\|$$
$$= 1.4\|\underline{w}\|^2$$

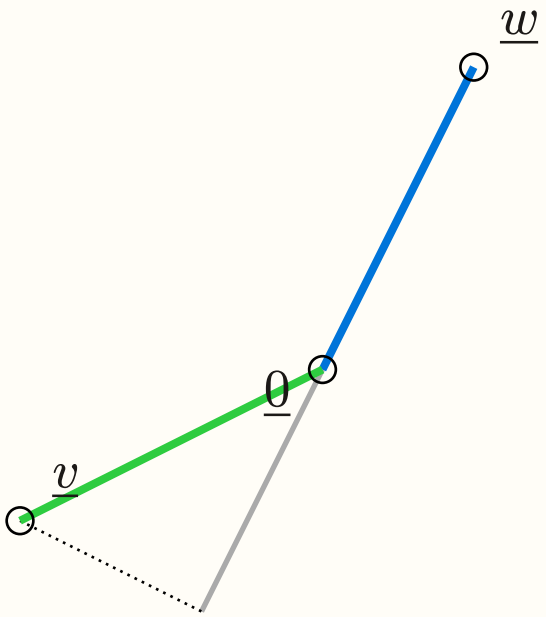
[1]

[1] remember that mathematicians irrationally hate the $\times$ sign!

Dot products as projections

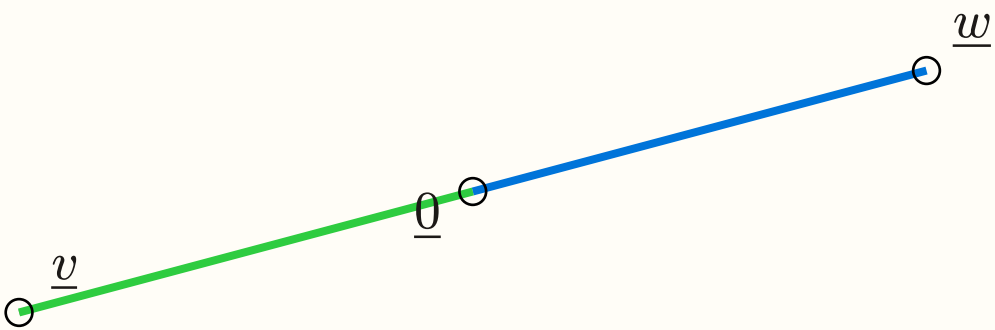


- $\langle \underline{v}, \underline{w} \rangle \approx 0.6 \|\underline{w}\|^2$



- $\langle \underline{v}, \underline{w} \rangle \approx -0.8 \|\underline{w}\|^2$
- went backwards from $\underline{0}$

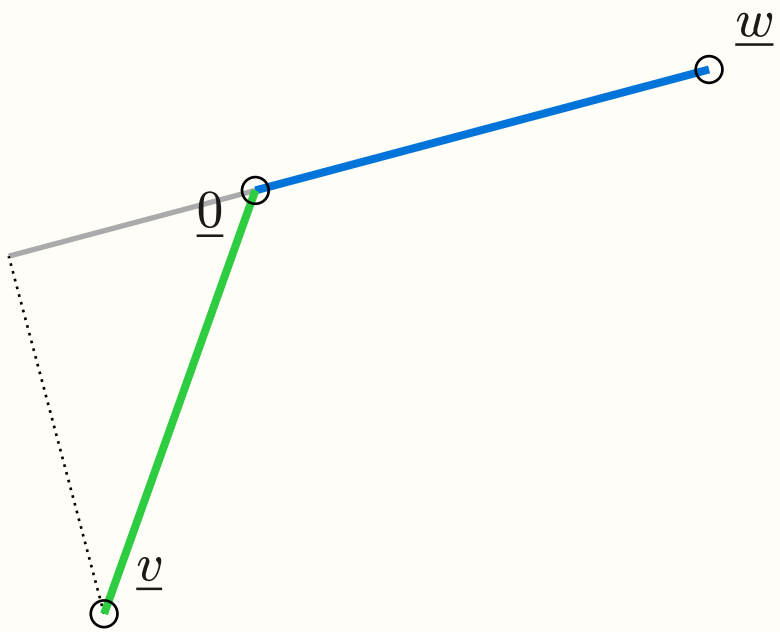
Dot products as projections



- Parallel in maths: $\underline{v} \propto \underline{w}$[1]
- $\langle \underline{v}, \underline{w} \rangle = k\|\underline{w}\|^2$

[1] $\propto$: proportional to. $\underline{v} = k\underline{w}$ for some scalar $k \in \mathbb{R}$

The Cauchy-Schwarz inequality



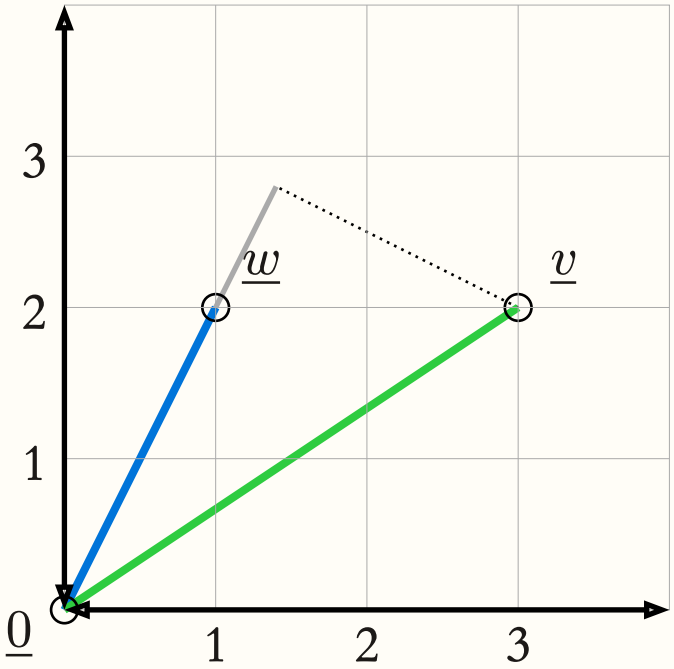
Shadow must have smaller norm than vector $\underline{v}$ itself

Exercise: prove from Pythagoras Theorem

Cauchy Schwarz

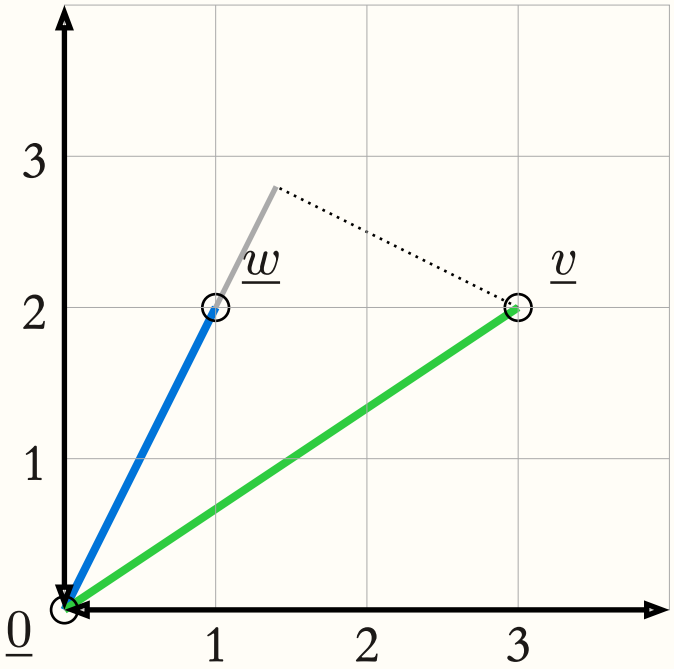
$$|\langle \underline{v}, \underline{w} \rangle| \leq \|\underline{v}\|_2 \|\underline{w}\|_2$$

Bilinearity of the dot product



$$\left\langle \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\rangle = 7$$

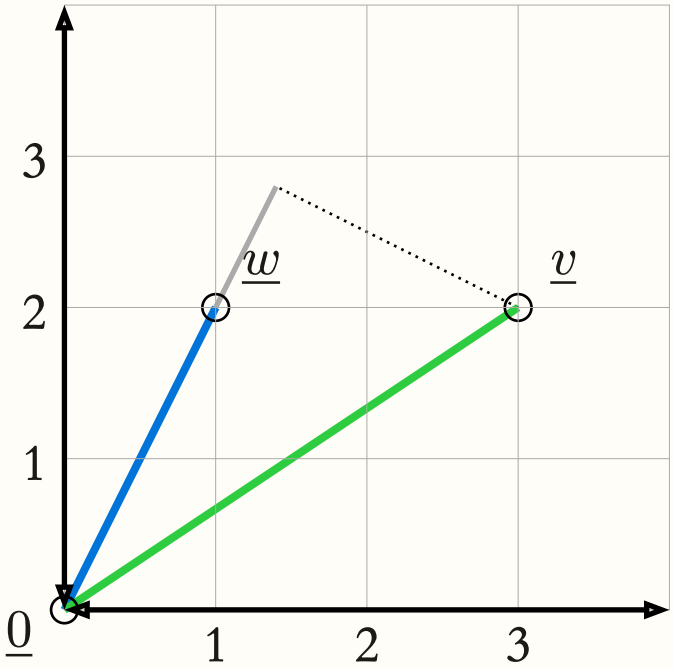
Bilinearity of the dot product



$$\left\langle \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\rangle = 7$$

$$\left\langle 2\begin{bmatrix} 3 \\ 2 \end{bmatrix}, 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\rangle ?$$

Bilinearity of the dot product



$$\left\langle \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\rangle = 7$$

$$\left\langle 2\begin{bmatrix} 3 \\ 2 \end{bmatrix}, 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\rangle ? = 2 \times 3 \times 7$$

Bilinearity in maths

$$\langle a\underline{v}, b\underline{w} \rangle = ab\langle \underline{v}, \underline{w} \rangle$$

$$\forall a, b \in \mathbb{R}, \underline{v}, \underline{w} \in V$$

Correlation

Definition

$$\operatorname{corr}(\underline{v}, \underline{w}) = \frac{\langle \underline{v}, \underline{w} \rangle}{\|\underline{v}\|_2 \|\underline{w}\|_2}$$

"Shrink onto unit circle, then take dot product"

Correlation

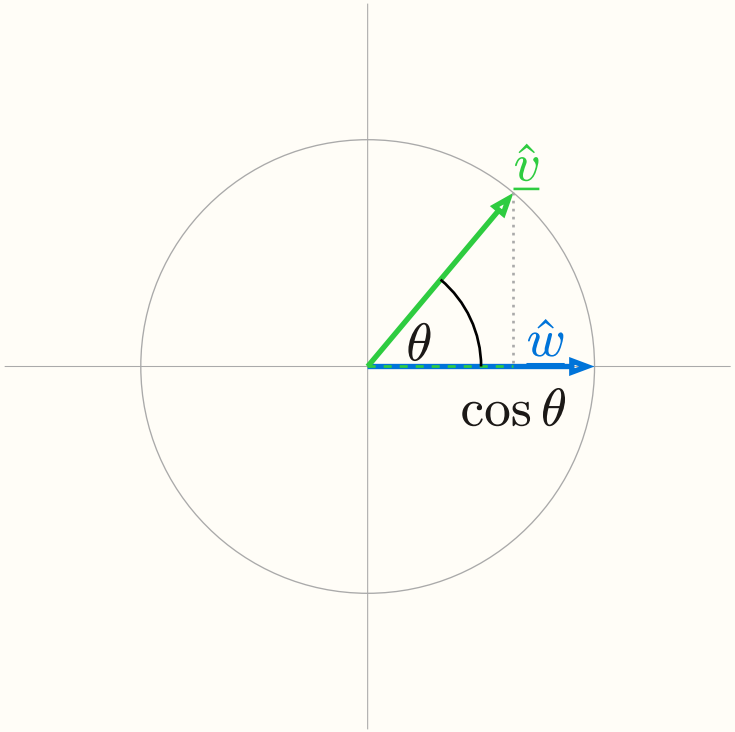
Definition

$$\operatorname{corr}(\underline{v}, \underline{w}) = \frac{\langle \underline{v}, \underline{w} \rangle}{\|\underline{v}\|_2 \|\underline{w}\|_2}$$

"Shrink onto unit circle, then take dot product"

Exercise

Prove that $\operatorname{corr}(\underline{v}, \underline{w}) = \cos(\theta)$ in $\mathbb{R}^2$



Next week: matrices as transformations